CN03: EXERCISE 1

Submission deadline: 11:59pm, 2026/10/17

- Please submit your report as one PDF file to our LMS site
- If you have questions on exercises, please ask the TA: d8272104@u-aizu.ac.jp

This exercise has two parts:

- *Lab questions*: simple network programming with Python. Your answer should have codes, screen captures, and discussions.
- *Homework questions*: related to the knowledge of the lectures 1~3.

Lab questions:

In this first exercise, you will write a simple Python program to connect to a web server. You may use the provided client template. Please answer the following questions:

1. In your client program, set ***serverName*** to be a web server. Run the client to connect to the server. Please show the reply message. (5 points)
   Hint: the reply message is in the variable **response**. Display the message by function print().
   Examples of web servers: www.yahoo.com, www.bbc.com, www.cnn.com, www.nhk.or.jp, etc.
2. What are the date, time, protocol, content type, and document type of the response? (5 points)
   Hint: look at the beginning part of the message.
3. Modify the program to compute and show the time delay of the program operation. Run the program 5 times and show the delays. Is there any difference in the time delays? Please explain. (15 points)
   Hint: at any point, the current time can be provided by the method time.time()
4. Some servers support both HTTP and HTTPS. Modify the code to compute the overhead of HTTPS. Is there any difference between the time delay of http and https? (5 points)
   Hint: replace "www" with "http://" or "https://" as appropriate
5. Now change the ***serverName*** to another web server located in Africa. Again, answer questions 2 and 3 with this new server. (10 points)
6. Discuss the differences of results between the two servers of questions 1 and 5. (5 points)
7. Compute the throughput of the connection in questions 1 and 5. (5 points)

Homework questions (each question = 5points)

Please answer the following questions:

1. Please give examples of protocols at each layer of Internet protocol stack
2. What are the roles of hosts, communication links, and packets switches?
3. What are routing and forwarding in networks?

4. Please explain the reasons for delay and loss on the Internet.
5. What are the differences between transport layer and network layer?
6. Please describe the relationships between process, socket, and port number.
7. What are addresses of a host and a process?
8. How can process A communicate to process B, which is on the same host, and to process C, which on a different host?
9. What are the differences between IPv4 header and IPv6 header?
10. What are the differences between transport services of UDP and TCP?